



## CAPSTONE PROJECT REGISTER

Class: \_\_\_\_\_ Duration time: from ....../.../.... To ....../... /.....

(\*) Profession: < Artificial Intelligence > Specialty: CV ☒ NLP ☐ DS ☐ Other ☐

(\*) Kinds of person make registers: Lecturer ☒ Students ☐

### 1. Register information for supervisor (if have)

|              | Full name          | Phone | E-Mail                                                   | Title       |
|--------------|--------------------|-------|----------------------------------------------------------|-------------|
| Supervisor 1 | Đặng Ngọc Minh Đức |       | <a href="mailto:ducdnm2@fe.edu.vn">ducdnm2@fe.edu.vn</a> | Assoc.Prof. |

### 2. Register information for students (if have)

|           | Full name | Student code | Phone | E-mail | Role in Group |
|-----------|-----------|--------------|-------|--------|---------------|
| Student 1 |           |              |       |        | Leader        |
| Student 2 |           |              |       |        | Member        |
| Student 3 |           |              |       |        | Member        |

### 3. Register content of the Capstone Project

(\*) 3.1. Capstone Project name:

English: Efficient Edge Vision Intelligence through Model Optimization

Vietnamese: Nghiên cứu hệ thống thị giác thông minh trên thiết bị biên thông qua tối ưu hóa mô hình

Abbreviation: EdgeVisionAI

#### a. **Context:**

Vision foundation models have greatly improved object detection and scene understanding, but most require substantial compute and are hard to deploy on resource-constrained edge devices. Edge AI enables intelligent processing directly on embedded devices, reducing latency, bandwidth, and privacy risk. Smart cities, intelligent transportation, industrial automation, and smart campuses increasingly rely on real-time visual intelligence near the data source. Lightweight vision models plus optimization frameworks such as TensorRT and ONNX Runtime now make it feasible to deploy advanced vision applications on platforms such as NVIDIA Jetson and Raspberry Pi AI boards.

**Research Gap:**

- Foundation models are too heavy for the edge: accurate vision models exceed the compute/memory/power budget of embedded devices.
- Accuracy–latency–energy trade-off: lightweight models trade accuracy for speed; the balance is hardware-specific and non-obvious.
- Optimizations interact unpredictably: quantization, TensorRT, distillation, and pruning behave differently per device, and systematic, reproducible benchmarking on real edge hardware is still limited.

Therefore, optimizing lightweight vision models to maximize real-time efficiency without unacceptable accuracy loss, benchmarked rigorously on fixed edge hardware, is an open, practically valuable research direction. This capstone targets a concrete, measurable technical contribution rather than a plain deployment demo.

**b. Objectives:**

The objectives of this capstone project are:

- Study Edge AI and embedded deep learning.
- Investigate recent lightweight vision / foundation models.
- Deploy computer-vision models on a fixed edge device.
- Optimize inference using compression and acceleration techniques.
- Evaluate real-time performance under different deployment configurations.
- Develop a prototype intelligent vision application.

**Research objective (contribution):**

Propose and experimentally validate at least one optimization strategy that measurably improves inference efficiency (FPS / latency / energy / memory) on the chosen edge device while keeping accuracy loss within an acceptable bound, supported by ablation and error analysis.

**c. Proposed solution:****Hardware:**

The target platform must be chosen and fixed in the first week and not changed mid-project:

- NVIDIA Jetson Nano.
- NVIDIA Jetson Orin Nano.
- Raspberry Pi (AI kit).

**Dataset Preparation:**

Use public datasets such as COCO and VisDrone (and optionally a small application-specific dataset for deployment evaluation). Standardize preprocessing and a fixed evaluation split.

### **Baseline Requirement:**

Each group must reproduce exactly one baseline (with a documented, unoptimized inference measurement on the fixed device) before any optimization:

- YOLO (an agreed, stable version — fixed at project start).
- RT-DETR.
- MobileNetV4.

### **Training Strategy (mandatory):**

Detectors are not trained from scratch. The pipeline is strictly: Pretrained Weights → Deployment → Optimization.

### **Required Improvement (Novelty):**

After the baseline is measured, students must apply at least one optimization (or an equivalent contribution) and quantify its effect:

- TensorRT optimization.
- FP16 optimization.
- INT8 quantization.
- Structured pruning.
- Knowledge distillation.
- ONNX Runtime optimization.

The improvement must be motivated by the Research Gap and evaluated against the unoptimized baseline on the same hardware.

### **Benchmark Protocol (standardized):**

All measurements follow a fixed procedure: (1) warm-up — discard the first 50–100 inferences; (2) then measure FPS, Latency, Memory, Model Size, and Power Consumption.

### **Experimental Evaluation:**

Report a Pareto analysis across Accuracy (mAP), FPS, Latency, Energy, and Memory, and benchmark each configuration at three precisions: FP32, FP16, and INT8.

### **Ablation study (required):**

- FP32 vs. FP16 vs. INT8.
- With vs. without TensorRT / ONNX Runtime.

- Different backbones (YOLO vs. RT-DETR vs. MobileNetV4).
- With vs. without knowledge distillation / pruning.

#### **Error analysis (required):**

- Failure cases (small objects, occlusion, motion blur, low light).
- Accuracy loss introduced by quantization/compression.
- Explicit accuracy–latency–energy trade-off (Pareto) discussion.

#### **Evaluation protocol (for consistency across groups):**

Report the exact device, input resolution, batch size, random seed(s), warm-up procedure, and how FPS/latency/energy/memory were measured.

#### **Prototype Development:**

A real-time intelligent vision application on the fixed embedded platform: live video processing, object detection, performance monitoring, and visualization of inference statistics (FPS, latency, memory, power).

#### **d. Algorithm requirement**

##### **Preparation**

- Datasets: COCO, VisDrone (and optional custom dataset).
- Models: one reproduced lightweight baseline (YOLO / RT-DETR / MobileNetV4) using pretrained weights only (no from-scratch training).
- Optimization: at least one of FP16 / INT8 / TensorRT / ONNX Runtime / distillation / pruning.

##### **Testing**

- Metrics: mAP, Precision, Recall, FPS, latency, memory, model size, power/energy.
- Baseline comparison: unoptimized baseline vs. optimized model(s); at least two deployment strategies; benchmark at FP32/FP16/INT8.
- Ablation and error analysis are mandatory.
- Evaluation protocol: report device, resolution, batch size, seed, warm-up, and measurement method.

##### **Deployment**

- Runtime environment: the fixed embedded AI platform (Jetson Nano / Orin Nano / Raspberry Pi AI) with Python + PyTorch, plus TensorRT / ONNX Runtime.
- Interface: a real-time application with live detection and an inference-statistics dashboard.

(\*) 3.2. Main proposal content (including result and product)

a. **Theory and practice (document):**

Apply knowledge from AIL302m (AI), AIP391 (Machine Learning), DPL301m (Deep Learning), Computer Vision, Embedded Systems, and Software Engineering. Literature review: Edge AI, Embedded Deep Learning, Real-time Object Detection, Vision Foundation Models, Lightweight CNN/Transformer architectures, Model Compression, Knowledge Distillation, Quantization, and TensorRT/ONNX optimization.

- Artificial Intelligence technologies:

Framework: PyTorch; TensorRT and ONNX Runtime for optimization/deployment.

Models: YOLO, RT-DETR, MobileNetV4, EfficientViT; foundation model reference: Florence-2.

Hardware: NVIDIA Jetson / Raspberry Pi AI; deployment in Python.

b. **Products:**

- A real-time Edge AI vision system on a fixed embedded platform.
- Optimized inference models (e.g., TensorRT/ONNX engines) at FP32/FP16/INT8.
- Experimental performance reports (ablation studies + Pareto trade-off analysis).
- A complete capstone thesis; results encouraged toward undergraduate research competitions or publications.

**Reproducibility package (required):**

Students must provide:

- Source code.
- Optimized model artifacts.
- Configuration files.
- Inference/benchmark script.
- README with setup and run instructions.
- Experiment log (metrics, hardware, seeds, settings per run).

Proposed Tasks (aligned with the AIP491 15-week schedule):

- **Task package 1 (Weeks 1–3):** Literature review, dataset preparation, hardware selection, environment setup, baseline model selection (→ Report 1: Introduction; Report 3: Existing Systems).
- **Task package 2 (Weeks 4–6):** Baseline implementation and initial (unoptimized) edge deployment with measured performance (→ Report 4: Methodology).

- **Task package 3 (Weeks 7–10):** Model optimization, edge deployment, performance tuning (the proposed improvement) (→ Report 4/5).
- **Task package 4 (Weeks 11–13):** Benchmarking (FP32/FP16/INT8), ablation studies, error/trade-off analysis, comparative analysis (→ Report 6 & 7: Results/Discussion).
- **Task package 5 (Weeks 14–15):** Prototype completion, thesis writing, and final presentation (→ Final report and defense).

### **AI Ethics and Responsible AI:**

Consistent with AIP491 CLO7 and the AI Ethics and Data Privacy session:

- Privacy — on-device processing reduces data exposure, but camera-based systems still raise surveillance and consent concerns.
- Bias and fairness — detection accuracy can vary across object types, lighting, and demographic contexts.
- Safety and reliability — consequences of missed/false detections in transportation or industrial settings.
- Responsible deployment — transparency about capabilities, limits, and appropriate human oversight.

### **Expected Learning Outcomes:**

Upon completion, students will be able to (mapped to AIP491 CLOs):

- Deploy and optimize deep-learning vision models on embedded/edge hardware (CLO3, CLO8).
- Reproduce a lightweight baseline, then propose and validate an optimization strategy (CLO3, CLO8, CLO10).
- Conduct reproducible benchmarks with proper protocol, ablation, and error analysis (CLO2, CLO5, CLO9).
- Critically analyze accuracy–latency–energy trade-offs (Pareto) (CLO9).
- Develop and deploy a real-time interactive prototype (CLO3, CLO6).
- Apply professional and ethical standards to camera-based AI systems (CLO7).
- Communicate research findings effectively in a written thesis and oral defense (CLO5, CLO6).

### **References:**

### **Foundations**

[1] B. Xiao, et al. Florence-2: Advancing a Unified Representation for a Variety of Vision Tasks. CVPR, 2024. (vision foundation model)

[2] J. Redmon, et al. You Only Look Once: Unified, Real-Time Object Detection. CVPR, 2016.

### **Benchmark / Dataset**

[3] T.-Y. Lin, et al. Microsoft COCO: Common Objects in Context. ECCV, 2014.

[4] P. Zhu, et al. Detection and Tracking Meet Drones Challenge (VisDrone). IEEE TPAMI, 2021.

### **Baseline Models**

[5] Y. Zhao, W. Lv, et al. DETRs Beat YOLOs on Real-time Object Detection (RT-DETR). CVPR, 2024.

[6] D. Qin, ..., A. Howard, et al. MobileNetV4: Universal Models for the Mobile Ecosystem. ECCV, 2024.

[7] H. Cai, et al. EfficientViT: Multi-Scale Linear Attention for High-Resolution Dense Prediction. ICCV, 2023.

### **Recent Representative Methods (Optimization)**

[8] B. Jacob, et al. Quantization and Training of Neural Networks for Efficient Integer-Arithmetic-Only Inference. CVPR, 2018.

[9] S. Han, H. Mao, and W. Dally. Deep Compression: Compressing Deep Neural Networks with Pruning, Trained Quantization and Huffman Coding. ICLR, 2016.

[10] NVIDIA. TensorRT Developer Guide, 2025; and ONNX Runtime (Microsoft), onnxruntime.ai. (deployment/optimization tools)

### **Survey**

[11] Z. Xu, et al. Edge Intelligence: Concepts, Architectures, Applications, and Future Directions. ACM Transactions on Embedded Computing Systems, 2021.

[12] J. Gou, et al. Knowledge Distillation: A Survey. International Journal of Computer Vision, 2021.

4. Other comments (propose all relevant things if any)

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**Supervisor (If have)**  
(Sign and full name)

**On behalf of the Registers**  
(Sign and full name)